

ESTIMATED NEW CANCER CASES AND DEATHS IN THE UNITED STATES 2000

Site of Origin	New Cases*		Deaths*	
	Male	Female	Male	Female
Breast	1,400	182,800	400	40,800
Colorectal	63,600	66,600	27,800	28,500
Esophagus	9,200	3,100	9,200	2,900
Kidney & Bladder	57,100	27,300	15,400	8,700
Leukemia	16,900	13,900	12,100	9,600
Liver	10,000	5,300	8,500	5,300
Lung	89,500	74,600	89,300	67,600
Lymphoid	35,900	26,400	14,400	13,100
Ovary	-	23,100	-	14,000
Pancreas	13,700	14,600	13,700	14,500
Prostate	180,400	-	31,900	-
Skin	34,100	22,800	6,000	3,600
Stomach	13,400	8,100	7,600	5,400
Testis	6,900	-	300	-
Uterine	-	48,900	-	11,100

*(the American Cancer Society's Clinical Oncology, Lenhard R.E., Osteen R.T., Gansler T., 2001)

Estimated new cancer cases and deaths in the United States in 2000. Adapted from Lenhard, 2001.

Benign or Malignant Tumor

Whether a tumor is benign or malignant determines how potentially life-threatening it is. Benign tumors are usually harmless, although their location may be serious (if surgery to remove the tumor would carry significant risk). These tumors are not considered cancerous, are relatively slow-growing, and usually are encased within a fibrous capsule.

Malignant tumors (cancers) have great potential to spread, or metastasize, to other sites in the body. These tumors are fast-growing and aggressive, and they invade neighboring healthy tissue. They therefore are considered life threatening.

Type of Tissue

The body consists of many different organs, which in turn are composed of several different types of tissues. There are three major categories of tissue-related tumor types: carcinoma, sarcoma, and leukemia/lymphoma. There are also other specialized tumor categories, such as those of the central nervous system (e.g., brain tumors).

Carcinoma. This is the largest category, containing about 90 percent of all cancers, and it consists of neoplasms derived from epithelial cells. Epithelial cells make up the outer layers of the skin. They also line the inner structures of organs such as the lungs, intestines and testes, as well as complex tissue such as the breast.

Sarcoma. These are solid tumors derived from all connective tissues except the blood-forming tissues (these are the leukemias and lymphomas). These tumors account for about 2 percent of all cancers. They occur in such tissues as muscle, bone, and cartilage.

Leukemia and Lymphoma. This group contains about 8 percent of all cancers, including blood cancers that originate from the marrow (leukemias) and from the lymphatic system (lymphomas). This group also includes other nonsolid tumors of the bone marrow and lymphatic system, such as myeloma, which affects plasma cells—a type of white blood cell found in the marrow and in other tissues.



Type of Cell

Classifying a tumor by the type of cell from which it is derived is slightly more complex than classifying it by the type of tissue, since there are so many cell types. The main cell types include adenomatous cells (which are ductal or glandular cells), basal cells (found at the base of the skin), myeloid blood cells (granulocytes, monocytes, and platelets), lymphoid cells (lymphocytes or macrophages), and squamous cells (flat cells). Therefore it is possible for a cancer classified by its site of origin to be broken up into one of several cell types. For example, a skin cancer could be either a squamous cell carcinoma, a basal cell carcinoma, or a melanoma (from a pigment-producing cell).

Site of Origin

Solid tumors are firm masses that develop from a neoplasm's originating organ, such as the brain, esophagus, kidney, liver, lung, ovary, pancreas, prostate, or testis. Tumors of the blood-forming tissues and lymphatic systems are not solid and tend to remain free and circulating even when malignant. Some of the common forms of cancer are listed in the table above.

Cancer Progression

There two main steps in cancer progression: the initial growth of the cancer and the subsequent spread via **metastasis**. Solid tumors are subject to the physiological constraints of biological systems: Without nutrients and oxygen, they will die. Therefore a solid tumor is initially limited in size to no larger than 1 to 2 millimeters in diameter (about the size of a small pea).

For a tumor to become aggressive, it needs to be able to nourish the cells at the center of its mass that are too far away from blood vessels. This is achieved by **angiogenesis**. Through mutation, a few cancer cells may gain the ability to produce angiogenic growth factors. These growth factors are proteins that are released by the tumor into nearby tissues, where they stimulate new blood vessels to grow into the tumor. This allows the tumor to rapidly expand in mass and invade surrounding tissue. It also provides a route for the cancer cells to escape into the new blood vessels and circulate throughout the body, where they can lodge in other organs forming metastases.

The most common way for a cancer to metastasize is through the lymphatic system. The lymphatic system is a network of channels throughout the body that carry a tissue fluid called lymph.

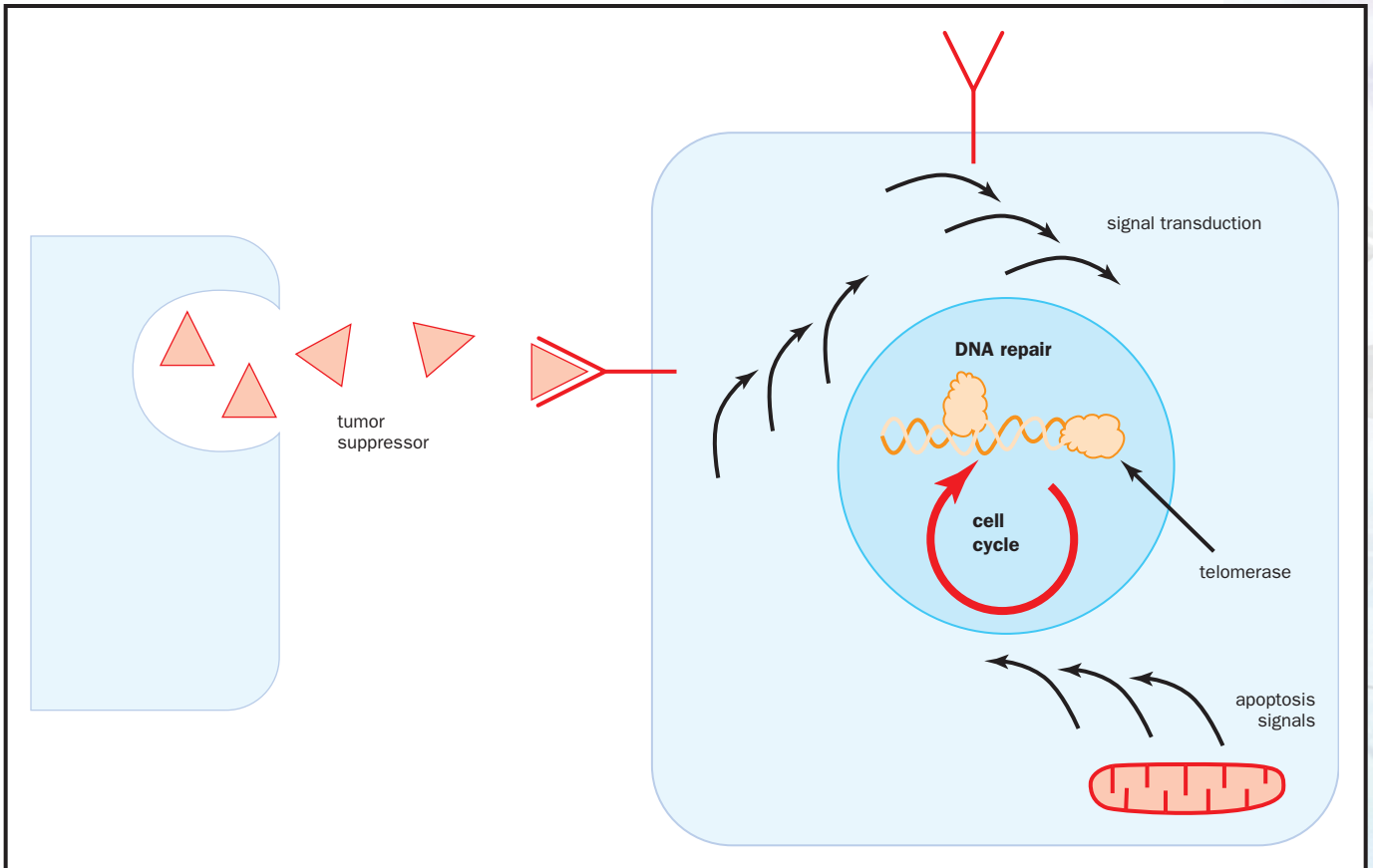
When a primary neoplasm metastasizes to another location, its cell type does not change. If leukemia metastasizes to the liver and develops a tumor, the tumor will display the characteristics of the leukemia, not those of a liver cancer. In some cases this can help physicians determine the original site of a tumor.

Genes Altered in Tumors

Although each cell in the body maintains itself and carries out its specific function, it is part of a large colony of collaborating cells that constitute the whole organism. A cell communicates with its surrounding cells by releasing chemical messages, in a process called signal transduction. These

metastasis breaking away of cancerous cells from the initial tumor

angiogenesis growth of new blood vessels



messages bind to specific receptor proteins on the surface of the surrounding cells. The gene expression of these cells is changed as a result of the messages.

A **hyperplastic cell** or a cancerous cell will stimulate neighboring cells to grow by secreting growth factors. Several types of genes can be mutated in tumor cells: oncogenes, tumor suppressor genes, DNA repair genes, and genes involved in cell mortality.

Oncogenes. These genes are involved in signal transduction, and some are involved in the various phases of the cell cycle. Mutations in cell-cycle regulation or signal transduction can “push” the cell into dividing rapidly and without regard to its surroundings. Over 100 oncogenes have been identified so far. They include genes such as *ABL1* (Abelson murine strain leukemia viral homolog) and *EGFR* (Epidermal Growth Factor Receptor).

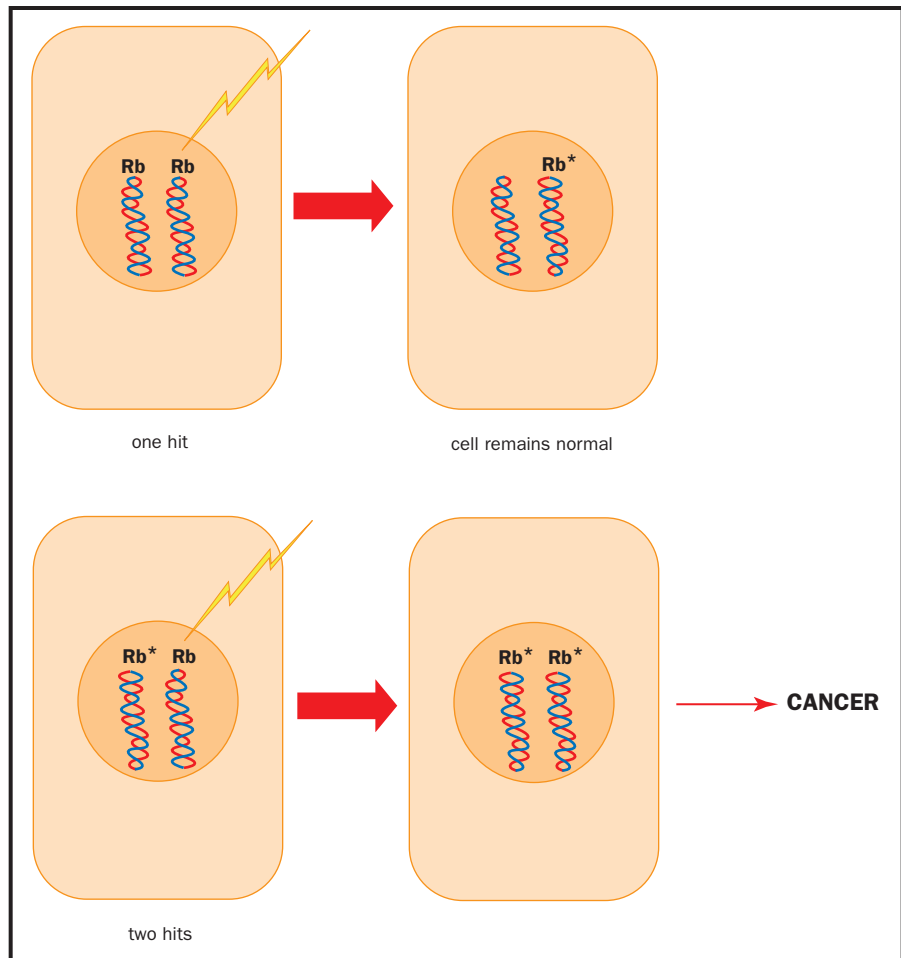
Tumor Suppressor Genes. These genes inhibit cell division, working in a manner opposite to that of the oncogenes. Surrounding cells secrete growth-inhibitory signals that help prevent proliferation. These growth-inhibitory signals work in conjunction with tumor suppressor genes. If a tumor suppressor gene is mutated, proliferating cells can ignore these inhibitory messages. This group includes the genes *p53*, *BRCA1*, and *BRCA2*.

DNA Repair Genes. These are the genes that provide the cell with the ability to sense and correct damage to the DNA. Damage to the DNA can be caused by radiation, chemicals, ultraviolet light, or errors in transcription. If these errors are not corrected, they accumulate in the genome and can

Multiple systems interact to control the cell cycle, ensuring that cell division occurs only when it is advantageous for the organism and when undamaged DNA is available for replication. Cancer may occur when any one of these systems is disrupted.

hyperplastic cell cell that is growing at an increased rate compared to normal cells, but is not yet cancerous

Inheritance of a mutated retinoblastoma gene (Rb^*) greatly increases the likelihood of developing the disease.



quickly increase the chance that a cell will become cancerous. Repair genes include those in the DNA-ligase and excision-repair gene families.

Genes Involved in Cell Mortality. A normal cell can only undergo about forty divisions, after which it dies or enters **senescence**. If a tumor had this limitation it would be very limited in its size, as it would reach its forty divisions relatively quickly. This process is controlled by the enzyme telomerase, which maintains the telomeres (repetitive DNA sequences at the ends of chromosomes that shorten after each round of DNA replication, until they reach a length that causes the cell to die) by not allowing them to shorten. Some cancer cells become immortal as a result of mutations in the telomerase gene, causing the telomeres to be extended indefinitely, allowing the cell to continue dividing without limit. Other mutations affect the process of **apoptosis**.

Cancer does not usually arise by a single event. Instead, two or more “hits” are needed to convert a well-regulated cell to a cancer cell. This is the case because each cell contains two copies of each gene, one inherited from each parent. Most cancer-causing mutations cause a loss of function in the mutated gene. Often, having only one functional copy is enough to prevent disease. Thus, two mutations are needed.

This can be illustrated by looking at retinoblastoma, a common cancer of the retina. The affected gene (called the *retinoblastoma* gene) is a tumor

senescence a state in a cell in which it will not divide again, even in the presence of growth factors

apoptosis programmed cell death

suppressor. **Spontaneous** mutations are rare, but since there are many millions of cells in the retina, several will develop the appropriate gene mutation over the course of a lifetime. It would be very unlikely, though, for a single cell to develop two spontaneous mutations (at least in the absence of prolonged exposure to carcinogens), and thus spontaneous retinoblastoma is very rare.

If, however, a person inherits one copy of an already-mutated gene from one parent, every cell in the eye starts life with one “hit.” The chances are very high that several cells will suffer another hit sometime during their life, and so the chances are very high that the person will develop retinoblastoma. Since inheriting a single copy of the mutated gene is so likely to lead to the disease, the gene is said to show a dominant inheritance pattern.

Future Directions in Diagnosis and Treatment

The increased knowledge of cancer at the biochemical and genetic level has led to many advances toward better diagnosis and treatment of cancer, including the design of more specific drugs that are less toxic to normal tissue. This includes the use of antisense molecules, which are nucleic acid sequences that are complementary to the **mRNA** of a target gene. As the two sequences are complementary, they anneal and thus the mRNA is blocked from being translated into a protein, resulting in less of that particular protein being produced (such as growth factor receptors). Drugs specific in blocking angiogenesis are able to control the growth and spread of tumors, especially when used in combination with other treatments. **SEE ALSO** AMES TEST; ANTISENSE NUCLEOTIDES; APOPTOSIS; BREAST CANCER; CARCINOGENS; CELL CYCLE; COLON CANCER; DNA REPAIR; MUTAGEN; MUTATION; ONCOGENES; SIGNAL TRANSDUCTION; TELOMERE; TUMOR SUPPRESSOR GENES.

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Carcinogens

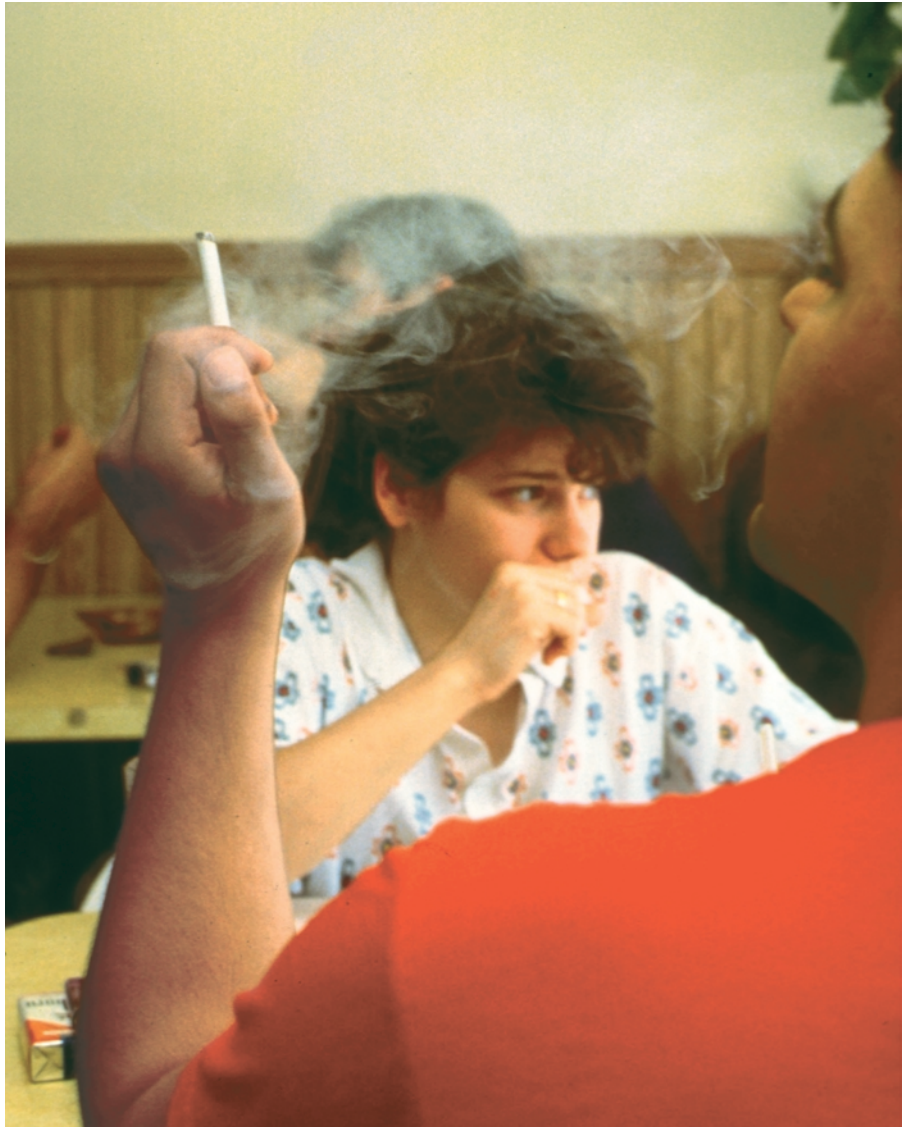
Carcinogens are agents that cause cancer, and include chemicals, radiation, and some viruses. While avoiding contact with carcinogenic agents is wise, it is virtually impossible to steer clear of them completely. Ultra-violet radiation from the sun, substances in food, and even oxygen can induce **malignancies**. In spite of the pervasive nature of carcinogens, however, not all individuals develop cancer, which suggests that mere contact

spontaneous non-inherited

mRNA messenger RNA

malignancies cancerous tissues

Breathing second hand smoke increases a person's heart rate and blood pressure, and delivers dangerous amounts of carbon monoxide to the blood. It is estimated that 2 percent of lung cancer deaths are caused by passive smoking.



with a carcinogenic agent is insufficient to produce this lethal disease. That is because organisms have evolved protective mechanisms to prevent cancer, and some of these defenses work by thwarting the potentially harmful effects of carcinogens.

Cancer-Causing Chemicals

References to cancer have been found in the annals of human disease since ancient times, but the disease's association with carcinogen exposure is a relatively new concept. Sir Percival Potts, a British physician who lived in the eighteenth century, was the first to suggest that the induction of cancer might be linked to agents in the environment. Potts had observed high rates of scrotal and nasal cancer among England's chimney sweeps, men who were exposed to accumulated fireplace soot during their work. After some careful studies, Potts suggested correctly that exposure to soot caused the high cancer rates, providing the impetus for identifying other carcinogens present in the environment.

In retrospect, it was fortuitous that soot was acknowledged as one of the first carcinogenic agents. Soot is a complex mixture of chemicals that arises from the combustion of organic material. As scientists and physicians separated soot's individual components, it became clear that chemicals called polycyclic aromatic hydrocarbons (PAHs) were among its principal carcinogenic compounds. The story became even more intriguing when it was shown that many PAHs behave as **procarcinogens**. Procarcinogens do not cause cancer per se, but they can be converted to active carcinogens by **enzymes** located in organs like the liver and lung. The implications of this discovery are noteworthy. For example, cigarette smoke contains a wide variety of procarcinogenic PAHs that are turned into active carcinogens in lung cells. Since smokers draw these PAHs deep into their lungs with each inhale on a cigarette, one reason that cigarette smoking correlates so highly with the induction of lung cancer becomes very clear.

Oncogenes and Tumor Suppressors

How do carcinogens cause cancer? Answering this question still forms the core of much basic research, but a common feature of many carcinogens, particularly chemicals and **radiation**, is that they act as mutagens. Mutagens are agents that generate changes in DNA, sometimes by reacting with the DNA building blocks, guanine, adenine, thymine, and cytosine, which results in damaged DNA. When such damage remains in chromosomes, genes are often mutated in a way that impairs their normal function and enhances cancer induction. Cells try to prevent such mutations by repairing DNA damage, but they are not always successful. In fact, some individuals are susceptible to hereditary skin and colon cancers because they lack the ability to remove damaged DNA from chromosomes.

There are two general classes of genes that contribute to malignant tumor formation when they are mutated by carcinogens: oncogenes and tumor suppressor genes. **Oncogenes** (the prefix “onco-” meaning “tumor”) are altered versions of normal genes called proto-oncogenes. Proto-oncogenes encode proteins that are often involved in regulating normal cell growth and division. When a proto-oncogene is mutated by exposure to a carcinogen, the protein it encodes may lose its ability to govern cell growth and division, often giving rise to the rapid, unrestrained cell proliferation that is characteristic of cancer. In such a case, the mutations in the proto-oncogene convert it into an actual oncogene.

While many oncogenes have been identified, numerous cancers are associated with mutations in one particular proto-oncogene, called *ras*, which is an abbreviation for “rat sarcoma.” “Ras” is written as Ras when biologists refer to the protein, and as *ras* when they refer to the gene that encodes the protein. The *ras* gene encodes Ras protein, which acts to regulate cell growth. Normally, Ras protein cycles between an “off” and “on” form. Many carcinogens induce **mutations** in the *ras* proto-oncogene, converting it to a *ras* oncogene, which encodes a form of the Ras protein that is locked in the “on” state. By abolishing Ras protein's regulatory off/on cycle, the accumulated mutations in the *ras* gene contribute to the formation of malignancies.

Not all oncogenes arise from mutations in normal cellular proto-oncogenes. In the early twentieth century, Peyton Rous discovered a

procarcinogens substances that can be converted into carcinogens, or cancer-causing substances

enzymes proteins that control a reaction in a cell

radiation high energy particles or waves capable of damaging DNA, including X rays and gamma rays

oncogenes genes that cause cancer

mutations changes in DNA sequences



carcinogenic virus that now bears his name, the Rous sarcoma virus. This virus harbors a gene called *v-src* (viral-sarcoma) that is a mutant form of a normal cellular proto-oncogene called *c-src* (cell-sarcoma). Like Ras protein, c-Src protein helps to regulate cell growth. When cells are infected by Rous sarcoma virus, the *v-src* gene, which is classified as an oncogene, is expressed in those cells. High amounts of mutant v-Src protein encoded by the *v-src* oncogene are made in the cell, and they dominate the normal cellular c-Src protein, an event that contributes to abnormal cell growth and proliferation, eventually leading to cancer.

Tumor suppressor genes encode proteins that tend to repress cancer formation. When tumor suppressor genes are mutated by carcinogens, they often lose their ability to stem tumor formation, resulting in cancer. Some hereditary forms of breast cancer are linked to mutations in a tumor suppressor gene called *BRCA-1*. BRCA is derived from BREast CAncer. The *BRCA-1* gene encodes BRCA-1 protein, which participates in controlling cell division, preventing cells from growing out of control, thus contributing to the suppression of tumor formation. Mutations in the *BRCA-1* gene result in altered BRCA-1 protein that no longer functions correctly in cell-growth regulation, contributing to the formation of tumors, particularly in breast tissue.

Reducing Exposure

Decreased carcinogen contact along with improved methods for treating cancer provide two important means for curtailing the suffering, expense, and death associated with the disease. The documented existence of carcinogens has prompted a worldwide effort to detect additional cancer-causing agents. A variety of toxicological assessments, including the Ames test, are used to identify potential mutagens and carcinogens. When possible, established carcinogens, such as asbestos, are removed from the environment, home, and workplace.

Exposure can also be reduced if the population is provided with protective warnings, like those advising the use of sunblock to shield skin from the cancer-causing effects of ultraviolet radiation in sunlight. The cost and manpower of such efforts are enormous, but carcinogen identification is critical for ensuring that exposure is minimized. A great challenge is reducing exposure to the carcinogens to which people actively expose themselves, most notably cigarette smoke. Prolonged education programs have helped cut down the use of cigarettes, but continued education is needed for each new generation. SEE ALSO AMES TEST; BREAST CANCER; CANCER; CELL CYCLE; DNA REPAIR; MUTAGEN; ONCOGENES; TUMOR SUPPRESSOR GENES.

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Cardiovascular Disease

Cardiovascular disease is a set of diseases affecting the heart and blood vessels. As with most chronic diseases whose incidence increases with age, it involves both inherited and environmental contributors and is therefore classified as a complex genetic disease. Most researchers believe that all major risk factors for cardiovascular disease have been identified. It is estimated that cigarette smoking, **hypertension**, abnormal serum cholesterol (low-density lipoprotein cholesterol or high-density lipoprotein cholesterol), obesity, lack of physical exercise, and diabetes account for 50 percent of the variability of risk in high-risk populations. The remaining risk is likely composed of a large number of yet-to-be identified minor risk factors or genetic influences that account for the development of disease in most individuals. Investigators who have attempted to estimate the overall contribution of genetics to the development of cardiovascular disease have proposed numbers ranging from 20 to 60 percent, based upon the analysis of large epidemiologic studies.

Finding Genes for Cardiovascular Disease

Genetics studies of cardiovascular disease involve searches for genes in two general classes: causative genes and disease-susceptibility (or disease-modifying) genes. These are sought through gene-linkage analysis or candidate-gene studies, respectively. Identifying causative genes for this disease is likely several years away at best. Before that time, however, a new understanding will have been reached regarding the relationship between inherited risks and outcomes in cardiovascular disease. With the development of new technology, we also have the promise of a detailed catalogue of disease-modifying genes that may open the door to therapeutic advances.

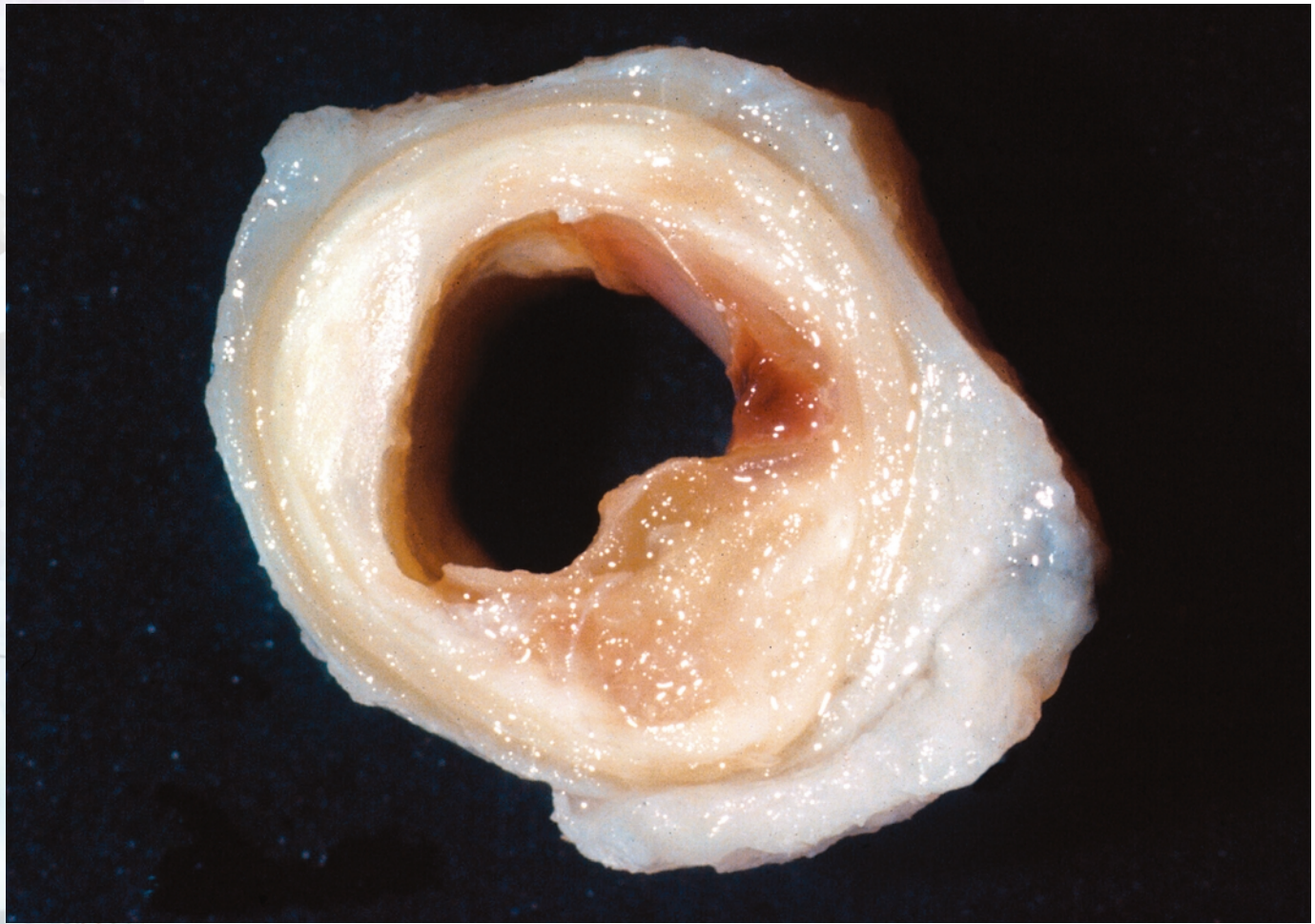
Gene-linkage analyses involve the study of families that express the cardiovascular trait of interest. In such studies, it is important also to establish the relative risk. Relative risk is defined as the probability of developing a condition (such as cardiovascular disease) if a risk factor (such as a gene) is present, divided by the probability of developing the condition if the risk factor is absent. A relative risk greater than 4.0 (that is, a four-fold greater risk due to presence of a gene or genes) will be associated with a reasonable likelihood of success in finding associated genes, given a study of 200 sibling pairs demonstrating the condition.

One of the best-studied types of cardiovascular disease is early-onset (or premature) coronary artery disease, which has a particularly strong genetic or inherited component. The coronary arteries are those around the heart that supply it with blood. Early-onset is defined as disease presentation (as reversible heart pain, heart attack, or cardiovascular surgery) before the age of 50. Approximately 8 to 10 percent of the U.S. population with cardiovascular disease presents before age 50, according to most surveys. Based upon a number of relatively small **epidemiologic** studies and several genetics studies in twins, a conservative estimate of the relative risk ratio contributed by genetics to the development of early-onset cardiovascular disease is between 4.0 and 8.0. Despite the fact that it has an inherited component, the actual genes responsible for familial predisposition to early-onset coronary artery disease have been incompletely investigated and remain obscure.

hypertension high blood pressure

epidemiologic the spread of diseases in a population





Coronary artery plaque is a symptom of cardiovascular disease. The buildup of excessive plaque can be attributed to both one's hereditary predisposition to coronary disease and one's lifestyle.

While population-level relative risk for developing cardiovascular disease can be known with a great deal of accuracy, therefore, this knowledge cannot be used to counsel or direct therapy for an individual in any given family. In fact, it has become clear to most practicing cardiologists that even when we know which cardiovascular risk factors are present, we have a very limited ability to predict the development of disease in most individuals.

Ongoing Studies

The ongoing studies of the genetics of cardiovascular disease consist of two general types: those that accumulate individual cases with the goal of performing association candidate-gene studies, and those that collect data from families (sibling pairs or extended families) with the idea of performing gene-linkage studies. Candidate-gene studies examine variations in genes that code for proteins that are likely to be involved in a disease or its prevention, such as genes controlling cholesterol metabolism or blood pressure. Linkage studies look for chromosome regions that are co-inherited with risk for disease, and then look carefully at the region to determine what genes are present.

Patients for both types of studies may be located in similar ways. Disease registry databases contain information on patients with particular conditions, which may have been collected by hospitals, charitable organizations, or research organizations. Clinical trials databases are generated during the

testing of a new drug or other treatment. Population-based longitudinal studies collect data on a large number of randomly selected people (not just those with disease) and follow them over many years, to determine what factors lead to development of disease. Each study has its own contribution to make, and only through the combined efforts of multiple studies and approaches will we discover and understand the genetic contributions to the development of cardiovascular disease.

Goals of Genetic Studies

Many of the promises of genetics investigations have probably been grossly overstated. The immediate potential of the ongoing and planned investigations into the genetics of cardiovascular disease is more promising for gene-directed therapy (the use of genetic information to guide the judicious use of medical interventions) than for somatic gene therapy (the use of a gene or gene product which, when introduced into a human organ, changes the function of the organ).

The realistic promises of current genetics studies include the elucidation of disease mechanisms; the identification of new targets for the development of therapeutic pharmacologic agents; and the use of genetic markers to identify individuals for whom a particular agent is either effective or unusually hazardous. This approach, called pharmacogenomics, improves the safety and efficacy of treatments, and enhances the ability to preferentially select subjects for clinical trials based upon genetic predisposition and for gene-directed therapy. In the latter case, for example, a genetic contributor to the development of early-onset cardiovascular disease might be used as an additional risk factor whose identification could focus the allocation of preventive resources, whether educational, behavioral, or pharmacologic, to populations at particularly high risk for the disease. **SEE ALSO** COMPLEX TRAITS; GENE AND ENVIRONMENT; PHARMACOGENETICS AND PHARMACOGENOMICS; PUBLIC HEALTH, GENETIC TECHNIQUES IN; STATISTICS.

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Cell Cycle

The cell cycle is the process by which a cell grows, duplicates its DNA, and divides into identical daughter cells. Cell cycle duration varies according to cell type and organism. In mammals, cell division occurs over a period of approximately twenty-four hours.

In multicellular organisms, only a subset of cells go through the cycle continuously. Those cells include the **stem cells** of the **hematopoietic** system, the basal cells of the skin, and the cells in the bottom of the **colon crypts**. Other cells, such as those that make up the endocrine glands, as well as liver cells, certain renal (kidney) tubular cells, and cells that belong to connective tissue, exist in a nonreplicating state but can enter the cell cycle after receiving signals from external stimuli. Finally, postmitotic cells are

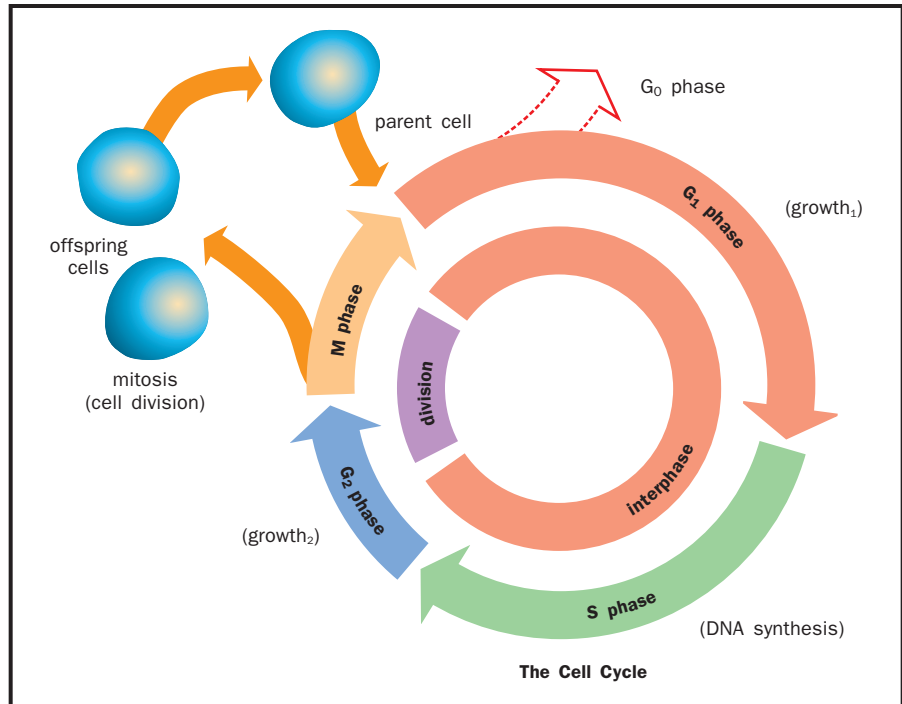
stem cells cells capable of differentiating into multiple other cell types

hematopoietic blood-forming

colon crypts part of the large intestine



Cyclin-dependent kinases (CDK) trigger transition between cell-cycle phases. Adapted from Robinson, 2001.



incapable of cell division even after maximal stimulation, and include most neurons, striated muscle cells, and heart muscle cells.

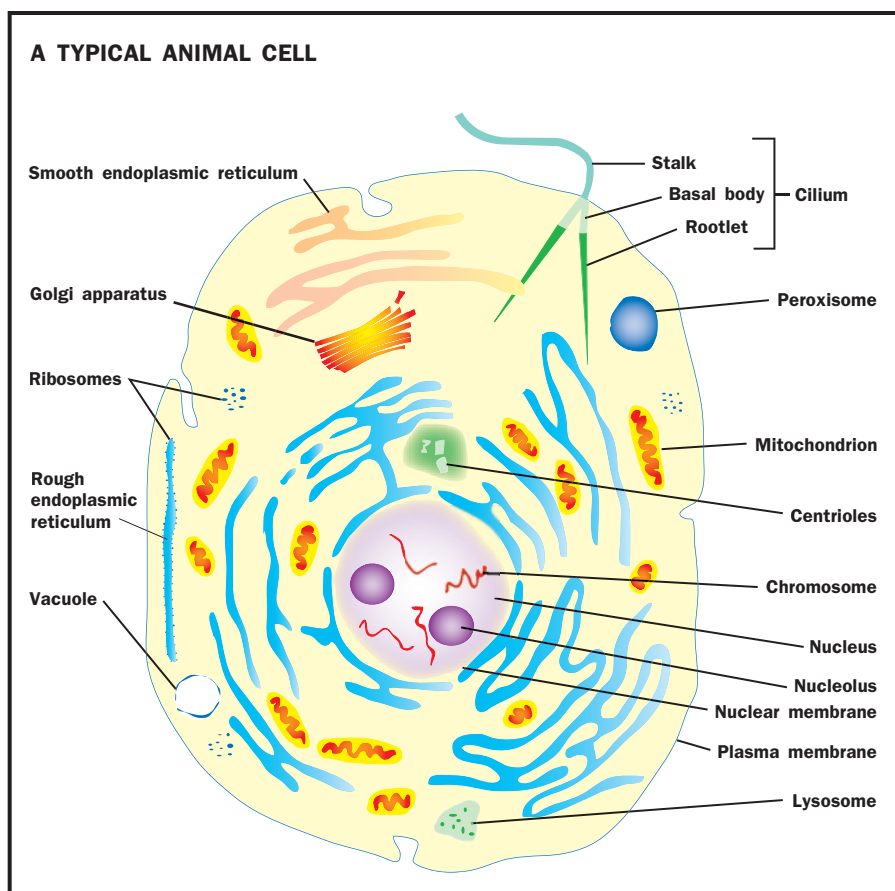
The cell cycle is functionally divided into discrete phases. During the DNA synthesis (S) phase, the cell replicates its chromosomes. During the mitosis (M) phase, the duplicated chromosomes are segregated, migrating to opposite poles of the cell. The cell then divides into two daughter cells, each having the same genetic components as the parental cell. Mammalian cells undergo two gap, or growth, phases (G₁ and G₂). G₁ occurs prior to the S phase, and G₂ occurs before the M phase.

Control of the Cycle

During the G₁ and G₂ phases, cells grow and make sure that conditions are proper for DNA replication and cell division. During the G₁ phase, cells monitor their environment and determine if conditions, including the availability of nutrients, growth factors and hormones, justify DNA replication. The decision to initiate replication is made at a specific “checkpoint” in G₁ called the “restriction point.”

The processes of DNA **replication** and mitosis, and intervening events during the cell cycle, occur in a highly ordered and specific manner. A complex network of proteins ensures that these events occur at the proper times. Intracellular and extracellular signals block cell-cycle progression at checkpoints if certain events have not yet been completed. After the restriction point, the cell is committed to replicating its genome and dividing, completing one round of the cell cycle. If, prior to the restriction point, cells sense inadequate growth conditions or receive inhibitory signals from other cells, they enter G₀ (G-zero) phase, also called quiescence. In the G₀ phase, they are maintained for prolonged periods in a nondividing state. If cells sense such conditions after the restriction point, they complete the current

replication duplication of DNA



Animal cell parts.
Adapted from Robinson,
2001.

round of the cell cycle and exit to G_0 during the subsequent G_1 phase. The G_2 phase is shorter than G_1 , but it, too, consists of important mechanisms that control the completion and fidelity of DNA replication and that prepare the cell for entry into mitosis. Whereas some conditions cause cells to enter the G_0 phase, others trigger **apoptosis**. One such signal that may trigger apoptosis is if a cell's DNA has undergone significant damage.

After the restriction point, at the transition from the G_2 to the M phase, another checkpoint occurs. Mitosis is prevented if DNA damage has occurred or if genomic replication is not complete. The final key checkpoint occurs at the end of mitosis, when the cycle stops if chromosomes are not properly attached to the mitotic spindle.

Proteins That Regulate the Cycle

The mammalian cell cycle control system is regulated by a group of protein **kinases** called **cyclin-dependent kinases (CDKs)**. These proteins catalyze the attachment of **phosphate groups** to specific serine or threonine amino acids in a target protein. The phosphate groups alter the target protein's properties, such as its interaction with other proteins. (The alteration of protein activity by the attachment of phosphate groups occurs frequently in cells.)

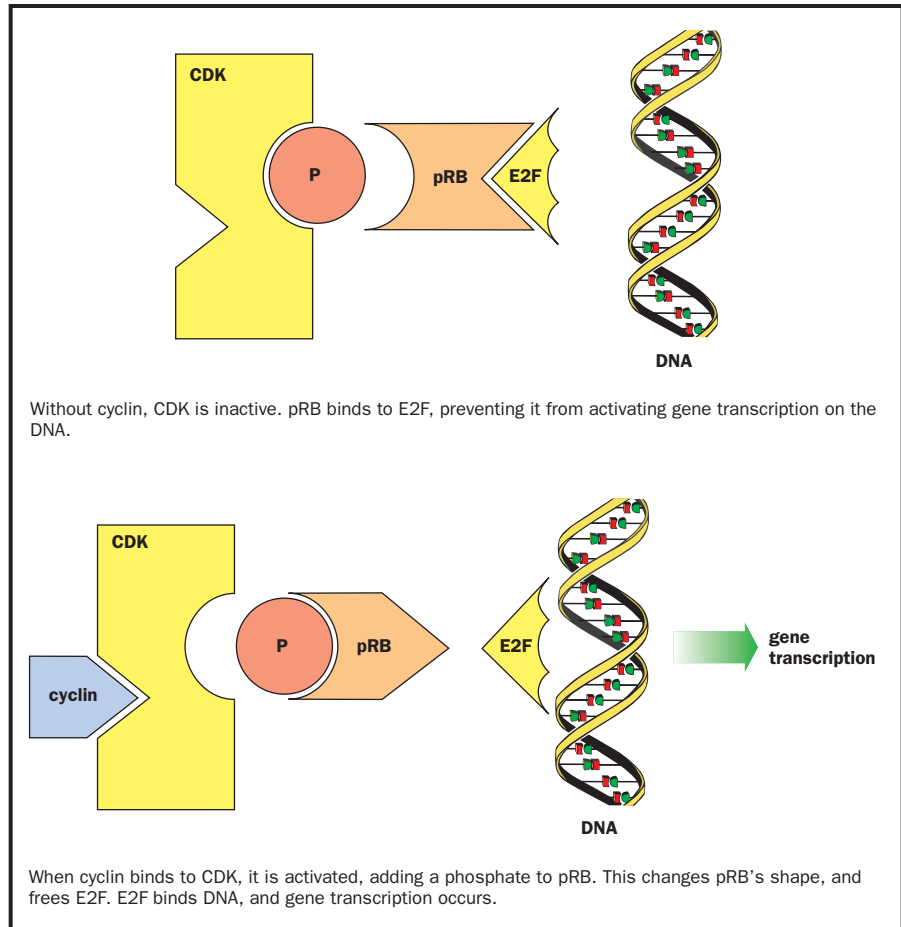
CDKs are called "cyclin-dependent" because their activity requires their association with activating subunits called cyclins. While the number of CDKs in a cell remains constant during the cell cycle, the levels of cyclins

apoptosis programmed
cell death

kinases enzymes that
add a phosphate group
to another molecule,
usually a protein

phosphate groups
 PO_4^{3-} groups, whose
presence or absence
often regulates protein
action

Figure 1.



oscillate. There are G_1 cyclins, S-phase cyclins, and G_2/M cyclins, each of which interact differently with CDK subunits to regulate the various phases of the cell cycle. CDKs can also associate with inhibitory subunits called CDK inhibitors (CKIs). In response to signals that work against proliferation, such as growth factor deprivation, DNA damage, cell-cell contact inhibition and lack of cell adhesion, CKIs cause the cell cycle to halt.

By the end of 2001, many structurally related cyclins (A1, A2, B1, B2, B3, B4, B5, C, D1, D2, D3, E1, E2, F, G1, G2, H, I, L, and T) and nine CDKs (CDK1 to CDK9) were identified in mammalian cells. Complexes of cyclin D and CDK4, as well as complexes of cyclin D and CDK6, operate during the G_1 phase. Complexes of cyclin A and CDK2, as well as complexes of cyclin E and CDK2, act during the transition from the G_1 to the S phase. Complexes of cyclin A and CDK1, as well as cyclin B and CDK1, function during the transition from the G_2 to the M phase.

Active complexes of cyclins and CDKs exert their biological effects by phosphorylating proteins. During the G_1 phase, a major target of cyclin/CDK complexes is the retinoblastoma protein (pRb). pRb is a growth-suppressing protein whose activity is controlled by whether or not it is phosphorylated.

When pRb is in the dephosphorylated form, during the G_0 phase and early in the G_1 phase, it is active. pRb exerts its growth-suppressing effects

by binding to many cellular proteins, including the **transcription factors** of the E2F family (Figure 1). E2F transcription factors regulate the expression of numerous genes that are expressed during G_1 , or at the transition from the G_1 to the S phase, to initiate DNA replication.

pRb that is bound to an E2F transcription factor inhibits the transcription factor's activity. Following phosphorylation by cyclin/CDK complexes, pRb dissociates from E2F, allowing the transcription factor to bind DNA sequences and activate the expression of genes necessary for the cell to enter the S phase. Cyclin D1/CDK4 complexes phosphorylation of pRb during the middle of the G_1 phase. They allow for subsequent phosphorylation of pRb by additional cyclin/CDK complexes that act later in the cell cycle.

Two families of CKIs have been identified, based on their **amino acid** sequence similarity and the specificity of their interactions with CDKs. One of the families of CKIs, the INK family, includes four proteins (p15, p16, p18 and p20). These CKIs exclusively bind complexes of cyclin D and CDK4, as well as complexes of cyclin D and CDK6, to block cells that are in the G_1 phase of the cell cycle. The other family of CKIs, the Cip/Kip family, consists of three proteins (p21, p27, and p57). These inhibitors bind to all complexes of cyclins and CDKs that function during the G_1 phase and during the transition from the G_1 to the S phase. They act preferentially, however, to block the activity of complexes containing CDK2.

Deregulation and Cancer

Deregulation of cell cycle control proteins plays a key role in the development of cancer. Overactivation of proteins that favor cell cycle progression, namely cyclins and CDKs, and the inactivation of proteins that impede cell cycle progression, such as CKIs, can result in uncontrolled cell proliferation.

In human **tumors**, it is genes encoding the proteins that control the transition from the G_1 to the S phase that are most commonly altered. These genes include those for cyclins, CKIs, and pRb. Such mutations overcome the inhibitory effects of pRb on the cell cycle, causing cells to have a growth advantage. In some cancers, this occurs after the direct mutation of the pRb gene, resulting in the protein's loss of function. In a larger set of cancers, pRb is indirectly inactivated by the hyper-activation of CDKs. This may result from over expression of cyclins, from an activating mutation in CDK4, or from inactivation of CKIs.

There is much evidence to suggest that cyclins can act as **oncogenes** to induce cells to become cancerous. In particular the G_1 cyclins, cyclin D1, and cyclin E have been implicated in the development of cancer. Overexpression of the cyclin D1 protein is frequently detected in human breast cancer, and increasing evidence suggests that cyclin E overexpression plays an important role in the **pathogenesis** of breast cancer.

CKIs antagonize the function of cyclins, and considerable evidence suggests that these proteins function as tumor suppressors. CKI function is often altered in cancer cells. The gene encoding p16, a protein that belongs to the INK family of CKIs, is mutated, deleted, or inactivated in a large number of human malignancies and tumors. Such alterations

transcription factors
proteins that increase the rate of gene transcription

amino acid a building block of protein

tumors masses of undifferentiated cells; may become cancerous

oncogenes genes that cause cancer

pathogenesis pathway leading to disease



prevent the inhibition of cyclin D/CDK4 and cyclin D/CDK6 complexes during G₁.

Decreased expression of p21 and p27, proteins that belong to the Cip/Kip family of CKIs, also has been demonstrated in numerous human tumors. In contrast to the genetic mutations observed with p16, the decrease in p27 levels in tumors is due to enhanced degradation of the p27 protein. One of the proteins required for the degradation of p27, Skp2, has oncogenic properties. Skp2 over expression is observed in several human cancers and likely contributes to the uncontrolled progression of the cell cycle by increasing the degradation of p27. Understanding of the fine details of cell cycle regulation is likely to lead to specific cancer therapies targeting one or more of these important proteins. SEE ALSO APOPTOSIS; CANCER; CELL, EUKARYOTIC; MEIOSIS; MITOSIS; ONCOGENES; REPLICATION; SIGNAL TRANSDUCTION; TUMOR SUPPRESSOR GENES; TRANSCRIPTION FACTORS.

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Cell, Eukaryotic

All living organisms are composed of cells. A eukaryotic cell is a cell with a nucleus, which contains the cell's chromosomes. Plants, animals, protists, and fungi have eukaryotic cells, unlike the Eubacteria and **Archaea**, whose cells do not have nuclei and are therefore termed prokaryotic. In addition to having a nucleus, eukaryotic cells differ from prokaryotic cells in being larger and much more structurally and functionally complex. Eukaryotic cells contain subcompartments called organelles, which carry out specialized reactions within their boundaries. A eukaryotic cell may be an individual organism, such as the amoeba, or a highly specialized part of a multicellular organism, such as a **neuron**.

Physical Characteristics

A typical eukaryotic cell is about 25 **micrometers** in diameter, but this average hides a large range of sizes. The smallest cell is a type of green algae, *Ostreococcus tauri*, with a diameter of only 0.8 micrometers, about the size of a typical bacterium. The human sperm is about 4 micrometers wide, but 40 micrometers long, while the egg is about 100 micrometers in diameter. Single neurons can be a meter or more in length. While schematic diagrams often picture cells as simple cubes or spheres, most cells have highly individual shapes. Human red blood cells are flattened disks indented on either

Archaea one of three domains of life, a type of cell without a nucleus

neuron nerve cell

micrometer 1/1000 of a meter

side; muscle cells are highly elongated; neurons are long and thin with many branches on each end; and white blood cells constantly change their shapes as they crawl through the body.

Cells are also often depicted as a bag of fluid with a smattering of structures within, but this is far from the truth. Instead, the interior of the cell is a dense network of structural proteins, collectively termed the cytoskeleton, within which is embedded a large collection of organelles. The material within the cell except for the nucleus is called the cytoplasm. The nonorganelle portion of the cytoplasm is called the cytosol. The consistency of the cytoplasm is much like egg white, and not at all like freely flowing water.

Membranes

Eukaryotic cells include large amounts of membrane, which enclose the cell itself and surround each of the organelles. The membrane surrounding the cell is termed the plasma membrane. Membranes are bilayered structures, made of two layers of phospholipid molecules, built from phosphoric acids and fatty acids. One end of the phospholipid molecules (the exterior head) is **hydrophilic**, and it is oriented to the outer side of the membrane; the other end (the interior tails) are **hydrophobic**. Despite this, water molecules can pass freely through the bilayer, as can oxygen and carbon dioxide. Ions such as sodium or chloride cannot pass through, however, and neither can larger molecules such as sugars or **amino acids**. Instead, these materials must pass through the membrane via specialized proteins. This selective permeability allows the membrane to control the flow of materials in and out of the cell and its organelles.

Proteins and Membrane Transport

Proteins are long chains of amino acids. They have unique shapes and chemical properties that dictate their diverse functions. Proteins govern the range of materials that enter and leave the cell, relay signals from the environment to the interior, and participate in many metabolic reactions, harvesting or harnessing energy to transform raw materials into the molecules needed by the cell for growth, repair, or other functions. Cytoskeleton proteins give the cell its structure. Approximately half the weight of a membrane is due to the proteins embedded in it. Proteins give each organelle, and the cell as a whole, its unique character.

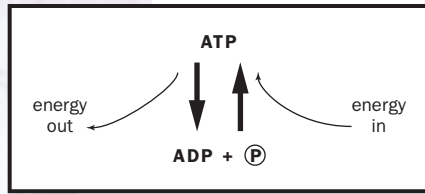
As noted, ions cannot pass freely through the cell's phospholipid membrane. Instead, most ions flow through special channels built from multiple protein subunits that together form a pore from one side of the membrane to the other. Some channels are gated, fitted with proteins that act as hinged doors, blocking the opening until stimulated to swing out of the way. Neurons, for instance, have gated sodium channels that open to allow an electrical impulse to pass and then close to recharge the cell for another firing. Molecules can also cross the membrane attached to protein pumps that are powered by **ATP**. Transport of scarce molecules such as sugars can also be powered indirectly, by coupling their movement to the flow of another substance. In addition to traversing the membrane directly, water passes through special channels formed by a protein called aquaporin.

hydrophilic water-loving

hydrophobic "water hating," such as oils

amino acids building blocks of protein

ATP adenosine triphosphate, a high-energy compound used to power cell processes



Energy is released when ATP is broken down, and required when it is formed.

hormones molecules released by one cell to influence another

transcription factor protein that increases the rate of transcription of a gene

glucose sugar

Signal Transduction

Proteins, including membrane proteins, also play critical roles in signal transduction, or relay. Signals can include **hormones**, ions, environmental changes such as odors or light, or mechanical disturbances such as stretching. A hormone is a small molecule released by one cell in the body to influence the behavior of another. A hormone exerts its influence by binding to a protein receptor in the target cell either on the membrane or within the cytoplasm. Cells that do not make receptors for a particular hormone are not susceptible to its effects. Adrenaline and testosterone are examples of hormones that illustrate two major modes of hormone action.

Adrenaline binds to a membrane-spanning receptor that projects both to the outside and the inside of the cell. The binding of adrenaline to the exterior portion changes the shape of the receptor, which in turn sets in motion other changes within the cell. The result is the production of a molecule called cyclic AMP (adenosine monophosphate), another form of the adenosine nucleotide. This “second messenger” binds to a variety of enzymes within the cell, activating them and leading to production of a variety of products. The exact set of enzymes turned on by cyclic AMP and the exact set of consequences depend on the particular cell. Kidney cells, for instance, increase their permeability to water, while liver cells release sugar into the bloodstream. The unique set of proteins within each cell is determined by the genes it has expressed, which in turn is determined by its own history and the hormones and other influences to which it has been exposed.

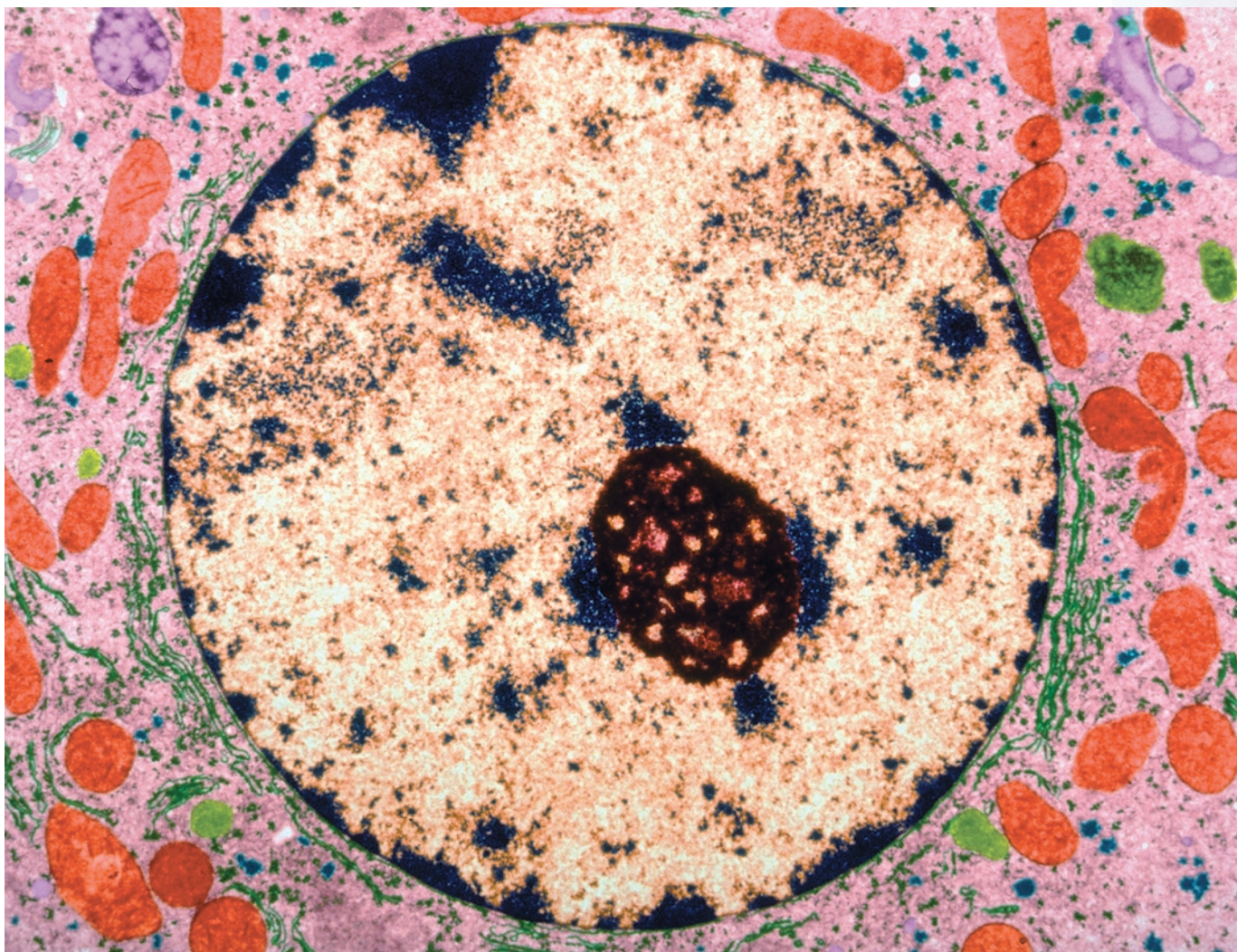
Testosterone’s effects come on more slowly than adrenaline’s, but last much longer. Testosterone passes through the plasma membrane and binds to a receptor in the cytosol. Once this occurs, the receptor-hormone complex is transported to the nucleus. Here, it binds to DNA, altering the rate of gene expression for a wide variety of genes. Thus, testosterone acts as a **transcription factor**. The prolonged action of testosterone is in part because it stimulates the production of new, long-lasting proteins that alter the cell’s function for much longer than the very rapid and short-lived effects of adrenaline.

Cells continually respond to signals, and they influence other cells through the signals they release. Signaling pathways within the cell control the rate of cell division, the development and differentiation of the cell, the secretion of proteins and other molecules, and the response to injury, among many other reactions.

Metabolism

Metabolism refers to the entire set of reactions within the cell. Most reactions can be classified as either anabolic or catabolic. Anabolic reactions use stored energy to build more complex molecules from simpler ones. Protein synthesis is an example. Catabolic reactions break down complex molecules to simpler ones, releasing energy in the process that may be harvested and stored by the cell. **Glucose** breakdown is an example.

The energy transfer in each type of reaction almost always involves the interconversion of ATP and ADP (adenosine diphosphate). Energy is released when ATP loses a phosphate to become ADP, while energy is required to make ATP from ADP and phosphate. ATP can also be con-



verted to AMP by the loss of two phosphates. This reaction, which releases even more energy, is used in replication of DNA and synthesis of RNA (transcription).

Mitochondrion

Glucose breakdown begins in the cytosol, but the majority of the process occurs in the mitochondrion, the energy-harvesting organelle of the cell. In addition to participating in the breakdown of glucose (and making ATP in the process), the mitochondrion is also involved in breaking down fats and amino acids. All these fuels are processed in two major steps, termed the Krebs cycle and the electron transport chain. In the Krebs cycle, the carbon skeletons are broken apart to make CO_2 , while the hydrogen atoms are removed on special nucleotide carriers. In the electron transport chain, the hydrogens are stripped of their energy in a series of steps to make ATP, and in the end are reacted with oxygen to form water. The mitochondrion consumes virtually all the oxygen used by the cell. The mitochondrion also participates in many anabolic reactions, using the intermediates of the Krebs cycle as a source of carbon skeletons for creating and modifying nucleotides, amino acids, and other building blocks of the cell.

The nucleus of this liver cell, magnified nearly 3,000 times, has been stained to show some of its components: DNA (dark purple), nucleolus (burgundy). Mitochondria (red), and cytoplasm (pink), are visible outside the nucleus.

The mitochondrion is the descendant of a once free-living bacterium that took up residence inside an ancient cell, probably to take advantage of high-energy molecules the host could not metabolize. Mitochondria retain their own DNA on their own bacteria-like chromosome, although over time most of the original mitochondrion's genes were transferred to the host and now reside in the nucleus.

Chloroplast

The cells of plants and some protists possess chloroplasts, whose green chlorophyll gives plant leaves their characteristic color. Embedded in an internal membrane, chlorophyll absorbs sunlight and funnels it to a complex set of proteins nearby. Light energy is used to split water into oxygen (released as a waste product) and hydrogen, which is attached to nucleotide carriers. The hydrogen is then reacted with CO₂ from the air to form sugars, the essential high-energy product that powers all of life. Like the mitochondrion, the chloroplast is a relic of a former free-living bacterium, and has its own DNA on its own chromosome.

Nucleus

The nucleus contains the chromosomes. Chromosomes contain the genes, which are DNA sequences used to create RNA. The nucleus is bounded by a double membrane, called the nuclear envelope. Numerous large pores provide channels through which materials enter and exit. One of the chief exports of the nucleus is messenger RNA, which is used in the cytoplasm for protein construction.

translation synthesis of protein using mRNA code

Translation occurs in the cytoplasm at ribosomes, large complexes made of protein and RNA. Ribosomes are assembled in the nucleus, in the region called the nucleolus. RNA is synthesized by the enzyme RNA polymerase, which unwinds DNA and transcribes short portions, known as genes. These RNA molecules are processed further before being exported as messenger RNA. Other RNAs made in the nucleus include the RNA used in ribosomes (ribosomal RNA), RNAs that carry amino acids to the ribosome (transfer RNA), and a host of small RNAs that mostly function in the nucleus to modify other RNAs.

Protein Synthesis, Modification, and Export

Messenger RNA exported from the nucleus binds to a ribosome in the **cytosol**, which then proceeds to translate the genetic message into a protein. Some proteins, with their ribosomes, remain free in the cytosol throughout translation, but others do not. Those that do not remain free carry a special sequence of amino acids at their leading end, called a signal peptide. This sequence directs the growing protein with its ribosome to the surface of the endoplasmic reticulum (ER), the most extensive organelle in the cell. Here, the ribosome attaches and extrudes the growing protein into the interior, or lumen, of the ER. Attachment of numerous ribosomes gives portions of the ER a rough appearance under the electron microscope. The ER also synthesizes most of the lipids used in the cell's many membranes. Lipid-synthesizing ER does not have ribosomes attached, and so appears smooth.

cytosol fluid portion of a cell, not including the organelles

Many of the proteins entering the ER lumen are destined for other compartments in the cell, and contain organelle-specific targeting sequences that

direct them to their final destination. Most of these proteins are first modified by the addition of branched sugar groups to make “glycoproteins.” Most proteins in the plasma membrane, for instance, are glycoproteins. The full range of functions of these sugar groups is unknown, but they may help the protein to fold correctly after synthesis, act in cell-cell recognition and adhesion, and promote appropriate interactions with other proteins.

Proteins are further modified and sorted in the Golgi apparatus, a set of flattened membrane disks that is continuous with the ER. Here proteins and lipids are packaged in **vesicles** that bud off and travel along the cytoskeleton to their final destination. Fusion of the vesicle membrane with the target membrane delivers the contents to the target organelle. Proteins and other materials that the cell exports travel to the plasma membrane via vesicles. Fusion of the vesicle with the **plasma membrane** delivers the contents to the exterior of the cell.

Cell Cycle

Cells must reproduce in order for the organism to grow or repair damage. For single-celled organisms, cellular reproduction creates a new organism. Each new cell must get a complete set of chromosomes, which therefore must be duplicated and evenly divided between the two daughter cells.

The orderly series of events involving cell growth and division is termed the cell cycle. Immediately following a division, the cell grows by taking up and metabolizing nutrients, and by synthesizing the many proteins, lipids, nucleic acids, sugars, and other molecules it needs. DNA replication occurs next, making duplicate chromosomes, followed by a short period in which the cell synthesizes the numerous proteins specific for cell division itself.

Cell division includes two linked processes: mitosis, or chromosome division, and **cytokinesis**, or cytoplasm division. Triggered by specific protein changes, the chromosomes begin to coil up tightly and become visible under the microscope. Cytoskeleton fibers attach to them, and position the chromosomes in pairs along the cell’s imaginary equator. At the same time, the nuclear envelope breaks down into numerous small vesicles. The cytoskeleton fibers (termed the spindle) pull the chromosome duplicates apart, segregating one member of each pair to opposite sides of the cell. Other cytoskeleton proteins pinch the membrane along the equator (in animal cells) or build a wall across it (in plant cells) to separate the two cell halves, ultimately forming two daughter cells. Finally, the nuclear envelope re-forms and the chromosomes uncoil, starting a new round of the cell cycle. SEE ALSO ARCHAEA; CELL CYCLE; EUBACTERIA; INHERITANCE, EXTRANUCLEAR; MEIOSIS; MITOCHONDRIAL GENOME; MITOSIS; NUCLEUS; PROTEINS; RIBOSOME; RNA PROCESSING; SIGNAL TRANSDUCTION; TRANSCRIPTION FACTORS; TRANSLATION.

Richard Robinson

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vesicles membrane-bound sacs

plasma membrane outer membrane of the cell

cytokinesis division of the cell’s cytoplasm



longitudinally length-wise

spindle football-shaped structure that separates chromosomes in mitosis

trilaminar three-layer

heterochromatin condensed portion of chromosomes

Centromere

During mitosis in a typical plant or animal cell, each chromosome divides **longitudinally** into two sister chromosomes that eventually separate and travel to opposite poles of the mitotic **spindle**. At the beginning of mitosis, when the sister chromosomes have split but are still paired, every chromosome attaches to the spindle at a specific point along its length. That point is referred to as the centromere or spindle attachment region.

Images from an electron microscope show that each sister is attached to fibers emanating from only one pole of the spindle. This allows the sisters to be pulled to opposite poles during mitosis. The electron microscope images also show that the spindle fibers do not terminate on the chromosomes themselves but rather on separate structures, known as kinetochores. Kinetochores are **trilaminar** bodies that assemble at the centromeres during the early stages of mitosis and disappear after the chromosomes have separated.

The budding yeast *Saccharomyces cerevisiae* has the simplest known centromeres consisting of a DNA segment only 110 bases in length. The DNA segment in the yeast centromere binds to specific proteins, which, like the kinetochores in higher organisms, link the chromosome to spindle fibers during mitosis. Centromeres of higher plants and animals are much larger, consisting of thousands or millions of bases of DNA and numerous proteins. For reasons that are unknown, centromeres are often flanked by long segments of DNA that do not contain functional genes. These nonfunctional DNA segments, called pericentric **heterochromatin**, vary in length in different organisms. In some cases they constitute more than half of the whole chromosome.

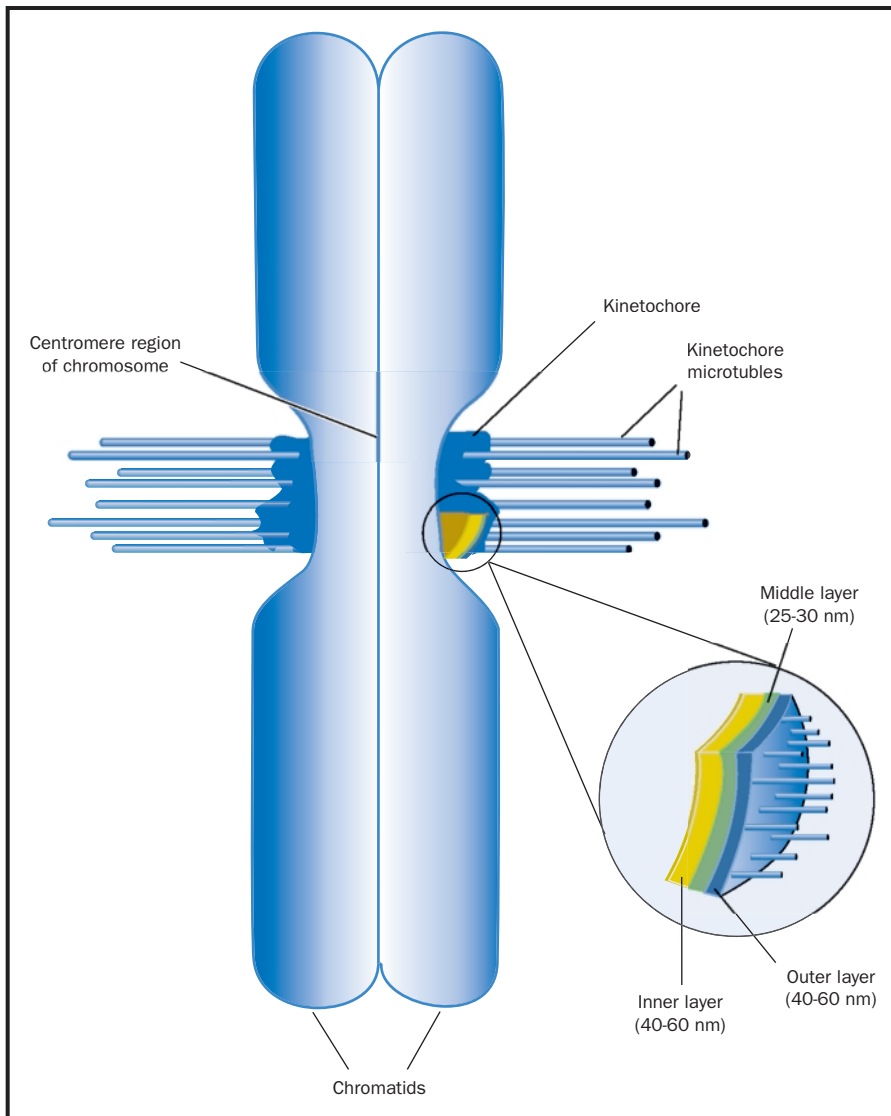
The crucial role of centromeres in the orderly behavior of chromosomes can be demonstrated by using X rays or other treatments that cause chromosomes to fragment. Pieces of chromosomes that lack centromeres (acentric fragments) do not attach to the spindle and are not pulled to the poles during mitosis. They generally are not included in the nuclei that are newly formed after cell division, and they usually degenerate in the cytoplasm.

Conversely, two fragments that each contain a centromere sometimes fuse, producing a dicentric chromosome. If the two centromeres happen to attach to the same pole at mitosis, the chromosome may move intact to that pole. However, if the centromeres attach to opposite poles, the chromosome will be stretched during mitosis and will eventually break. In general, therefore, only chromosomes with one centromere are stable.

Some organisms, including hemipteran insects and nematode worms, have *holocentric* or *holokinetic* chromosomes. In these organisms, spindle fibers attach all along one side of each sister chromosome, and the chromosomes are pulled more or less sideways to the pole.

Centromeres also play an important role during meiosis, in which the number of chromosomes is halved. The first meiotic division differs from a typical mitotic division in two respects:

1. In the first meiotic division, chromosomes derived from the organism's maternal and paternal parents pair at the beginning of meiosis. As in



The kinetochore is a protein structure that forms at the centromere before mitosis. Microtubules attach to it, and pull apart the two chromatids (not shown to scale). Adapted from <<http://www.sus.mcgill.ca/bio202/00b/lec2/sld028.htm> and <http://www.esb.utexas.edu/dr325/Supplements/kinetchr.htm>>.

a typical mitosis, each of these chromosomes has split into two sisters, so after pairing there are four chromosomes in a group.

2. When these four chromosomes attach to the spindle, sister chromosomes attach to the *same* pole, not to opposite poles as occurs in mitosis.

As a result, both maternal chromosomes move to one pole, while both paternal chromosomes move to the opposite pole. It is this unique behavior of the centromeres at meiosis that accounts for the separation of maternal and paternal genes during formation of sperm and eggs, which in turn is the basis of Mendelian genetics. SEE ALSO CELL CYCLE; CHROMOSOME, EUKARYOTIC; MEIOSIS; MENDELIAN GENETICS; MITOSIS.

Joseph G. Gall

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